



Risk of psychiatric disorders in Guillain-Barre syndrome: A nationwide, population-based, cohort study



Nian-Sheng Tzeng^{a,b}, Hsin-An Chang^{a,b}, Chi-Hsiang Chung^{c,d,e}, Fu-Huang Lin^d, Chin-Bin Yeh^{a,f}, San-Yuan Huang^{a,f}, Chuan-Chia Chang^a, Ru-Band Lu^{a,f,g,h,i,j,k}, Yu-Chen Kao^{a,l}, Hui-Wen Yeh^{a,m,n,o}, Wei-Shan Chiang^{a,p}, Wu-Chien Chien^{d,e,*}

^a Department of Psychiatry, Tri-Service General Hospital, School of Medicine, National Defense Medical Center, Taipei, Taiwan, ROC

^b Student Counseling Center, National Defense Medical Center, Taipei, Taiwan, ROC

^c Taiwanese Injury Prevention and Safety Promotion Association, Taipei, Taiwan, ROC

^d School of Public Health, National Defense Medical Center, Taipei, Taiwan, ROC

^e Department of Medical Research, Tri-Service General Hospital, National Defense Medical Center, Taipei, Taiwan, ROC

^f Graduate Institute of Medical Sciences, National Defense Medical Center, Taipei, Taiwan, ROC

^g Division of Clinical Psychology, Institute of Allied Health Sciences, College of Medicine, National Cheng Kung University, Tainan, Taiwan, ROC

^h Department of Psychiatry, College of Medicine, National Cheng Kung University, Tainan, Taiwan, ROC

ⁱ Institute of Behavioral Medicine, College of Medicine, National Cheng Kung University, Tainan, Taiwan, ROC

^j Department of Psychiatry, National Cheng Kung University Hospital, Tainan, Taiwan, ROC

^k Center for Neuropsychiatric Research, National Health Research Institute, Zhunan, Miaoli County, Taiwan, ROC

^l Department of Psychiatry, Tri-Service General Hospital, Song-Shan Branch, National Defense Medical Center, Taipei, Taiwan, ROC

^m Institute of Bioinformatics and Systems Biology, National Chia Tung University, Hsin-Chu, Taiwan, ROC

ⁿ Department of Nursing, Tri-Service General Hospital, School of Nursing, National Defense Medical Center, Taipei, Taiwan, ROC

^o Department of Nursing, Kang-Ning University (Taipei Campus), Taipei, Taiwan, ROC

^p Department and Institute of Mathematics, Tamkang University, New Taipei City, Taiwan, ROC

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ABSTRACT

Background: Guillain-Barre syndrome (GBS) is a rare immune-related neurological disorder with high mortality and morbidity, but the comorbid psychiatric disorders garnered little attention in the GBS patients. This study aimed to investigate the association between GBS and the risk of developing psychiatric disorders.

Methods: A total of 18,192 enrolled patients, with 4548 study subjects who had suffered GBS, and 13,644 controls matched for gender and age, from the Inpatient Dataset of 2000–2013 in Taiwan, and selected from the National Health Insurance Research Database (NHIRD). After adjusting for confounding factors, Cox proportional hazards analysis was used to compare the risk of developing psychiatric disorders during the 13 years of follow-up.

Results: Of the study subjects, 471 (10.35%) developed psychiatric disorders when compared to 1023 (7.50%) in the control group. Fine and Gray's competing risk model analysis revealed that the study subjects were more likely to develop psychiatric disorders (crude hazard ratio [HR]: 4.281 (95% CI = 3.819–4.798, $p < 0.001$). After adjusting for gender, age, monthly income, urbanization level, geographic region, and comorbidities, the adjusted HR was 4.320 (95% CI = 3.852–4.842, $p < 0.001$). Dementia, depressive disorders, sleep disorders, and psychotic disorders predominate in these psychiatric disorders. Mechanical ventilation and hemodialysis are associated with a lower risk of dementia when compared to the control groups.

Conclusions: Patients who suffered from GBS had a higher risk of developing psychiatric disorders, and this finding should act as a reminder to the clinicians that a regular psychiatric follow-up might well be needed for those patients.

* Corresponding author at: Department of Medical Research, Tri-Service General Hospital, National Defense Medical Center, 7115R, No. 325, Section 2, Cheng-Kung Road, Neihu District, Taipei City 11490, Taiwan, ROC.

E-mail address: chienwu@ndmctsgh.edu.tw (W.-C. Chien).

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1. Introduction

Guillain-Barre syndrome (GBS) is a rare immune-related neurological disorder [1,2], with a prevalence of around 1.7 per 100,000 person-years in Taiwan [3,4], which might well contribute to their morbidity or mortality [4–6]. Some articles about traumatic stress and psychological sequela [7–9] could also be associated with patients with GBS, but only from limited studies on anxiety, depression, sleep, and psychotic disorder in small sample studies or case reports [10–15], in comparison to another immune-related neurological disorder, multiple sclerosis, which has had more comprehensive studies about the prevalence of anxiety, alcohol abuse, bipolar disorder, depression, substance abuse, and psychotic disorders [16,17].

Psychiatric, or mental disorders, are defined as clinically significant behavioral or psychological syndromes, which are associated with present distress, disability, or an increased risk of suffering death, pain, or disability, and subsequent behavioral, psychological, or biological dysfunctions [18,19]. Previous studies have found that psychiatric disorders, such as dementia, anxiety disorders, depressive disorders, bipolar disorders, sleep disorders, psychotic disorders, and posttraumatic stress disorder or acute stress disorder, are associated with occupational injury [20], epilepsy [21], substance use disorders [22], and patients who have had weight control surgery [23]. Psychological stressors, or injury to the brain in the diseases, might contribute to these associations in injury, diseases, substance use disorders, or a post-surgery state. However, the association between GBS and psychiatric disorders has not, as yet, been studied. Therefore, a nationwide, population-based study is necessary for the association between GBS and the risk of psychiatric disorders for the clinicians who actually cared for these patients.

The National Health Insurance (NHI) Program was launched in Taiwan in 1995, and as of June 2009, it included contracts with 97% of the medical providers with approximately 23 million beneficiaries, or > 99% of the entire population [24,25]. The National Health Insurance Research Database (NHIRD), which contains all claims data of the beneficiaries, uses the International Classification of Diseases, 9th Revision, Clinical Modification (ICD-9-CM) codes to record diagnoses [26]. All diagnoses of psychiatric disorders in Taiwan are made by board-certified psychiatrists. An Inpatient Dataset in 2000–2013 was selected from the NHIRD, as several studies have demonstrated the accuracy and validity of the diagnoses [27–29]. Therefore, we used the NHIRD to study the association between GBS and the risk of psychiatric disorders.

2. Methods

2.1. Data sources

The present study used the NHIRD to identify inpatients with a discharge diagnosis of GBS, based on the International Classification of Diseases, 9th Revision, Clinical Modification (ICD-9-CM) code (357.0) during 2000–2013. The NHI Administration randomly reviews the records of ambulatory care visits and in-patient claims periodically to verify the accuracy of the diagnoses [30].

2.2. Study design and sampled participants

This study was of a population-based, matched-cohort design. Patients with newly diagnosed GBS were selected from the Inpatient Dataset from January 1, 2000, to December 31, 2013, according to a diagnosis of GBS (ICD-9-CM 357.0). The patients with GBS before 2000 were excluded. In addition, the patients diagnosed with dementia, depressive disorders, anxiety disorders, bipolar disorders, sleep disorders, psychotic disorders, posttraumatic stress disorder (PTSD), and acute stress disorder before 2000, or before the first visit for GBS, were also excluded. All patients aged < 20 were also excluded. All patients with

other neurological disorders who were admitted for conditions that were similar to GBS, such as chronic inflammatory demyelinating polyneuropathy (357.81), critical illness polyneuropathy (357.82), critical illness myopathy (359.81), polyneuropathy because of other diseases (e.g., porphyria and diphtheria; 357.4), acute poliomyelitis (045, 045.1), myasthenia gravis, other myasthenic syndrome (358.0, 358.01, 358.1, 358.8), acute transverse myelitis (323), acute alcohol intoxication (303.0), and poisoning by drug and biologic substances (960, 979) were excluded, with reference to one previous study related to GBS [4]. The patient's unique identifier was used to ensure that we only evaluated data for the first admission in cases with multiple admissions. A total of the patients that were enrolled, including 4548 subjects with GBS and 13,644 controls without GBS, were matched for age, gender, and index year (Fig. 1).

2.3. Ethics

This study was conducted in accordance with the Code of Ethics of the World Medical Association (Declaration of Helsinki). The Institutional Review Board of the Tri-Service General Hospital approved this study and waived the need for individual written informed consent (IRB No. 2-105-05-082).

2.4. Covariates

The covariates included gender, age groups (20–29, 30–39, 40–49, 50–59, 60–69, ≥ 70 years), geographical area of residence (north, center, south, and east of Taiwan), urbanization level of residence (levels 1 to 4), and monthly income (in New Taiwan Dollars [NT\$]; < 18,000, 18,000–34,999, ≥ 35,000). The urbanization level of residence was defined according to the population and various indicators of the level of development. Level 1 was defined as a population of > 1,250,000, and a specific designation as political, economic, cultural, and metropolitan development. Level 2 was defined as a population between 500,000 and 1,249,999, and as playing an important role in the politics, economy, and culture. Urbanization levels 3 and 4 were defined as a population between 149,999 and 499,999, and < 149,999, respectively [31]. Use of a mechanical ventilator was defined as ICD-9-CM procedure codes 96.7 (invasive mechanical ventilator), and 93.9 (non-invasive mechanical ventilator). Use of a therapeutic plasma exchange was defined as the ICD-9-CM procedure code 99.71.

2.5. Comorbidity

Comorbidities were assessed using the Charlson Comorbidity Index (CCI), which categorizes comorbidities using ICD-9-CM codes, scores each comorbidity category [1,2,32–35], and combines all scores to calculate a single comorbidity score. A score of zero indicates that no comorbidities were found, and higher scores indicate higher comorbidity burdens [25]. The five major complications that we evaluated were cardiac complications, pulmonary complications, respiratory failure necessitating endotracheal intubation, systemic infection, and thrombotic complications.

2.6. Outcome measures

All of the study participants were followed from the index date until the onset of dementia (ICD-9-CM codes: 290.0, 290.10, 290.11, 290.12, 290.13, 290.20, 290.21, 290.3, 290.41, 290.42, 290.43, 290.8, 290.9, and 331.0), anxiety disorders (ICD-9-CM 300), depressive disorders (ICD-9-CM 296.2–296.3, 300.4, 311), bipolar disorders (ICD-9-CM 296.0, 296.4–296.8), sleep disorders (ICD-9-CM 307.4, 780.5), psychotic disorders (ICD-9-CM 295, 297–298), and posttraumatic stress disorder (PTSD) and acute stress disorder (ICD-9-CM 308, 309.81), withdrawal from the NHI program, or the end of 2013.

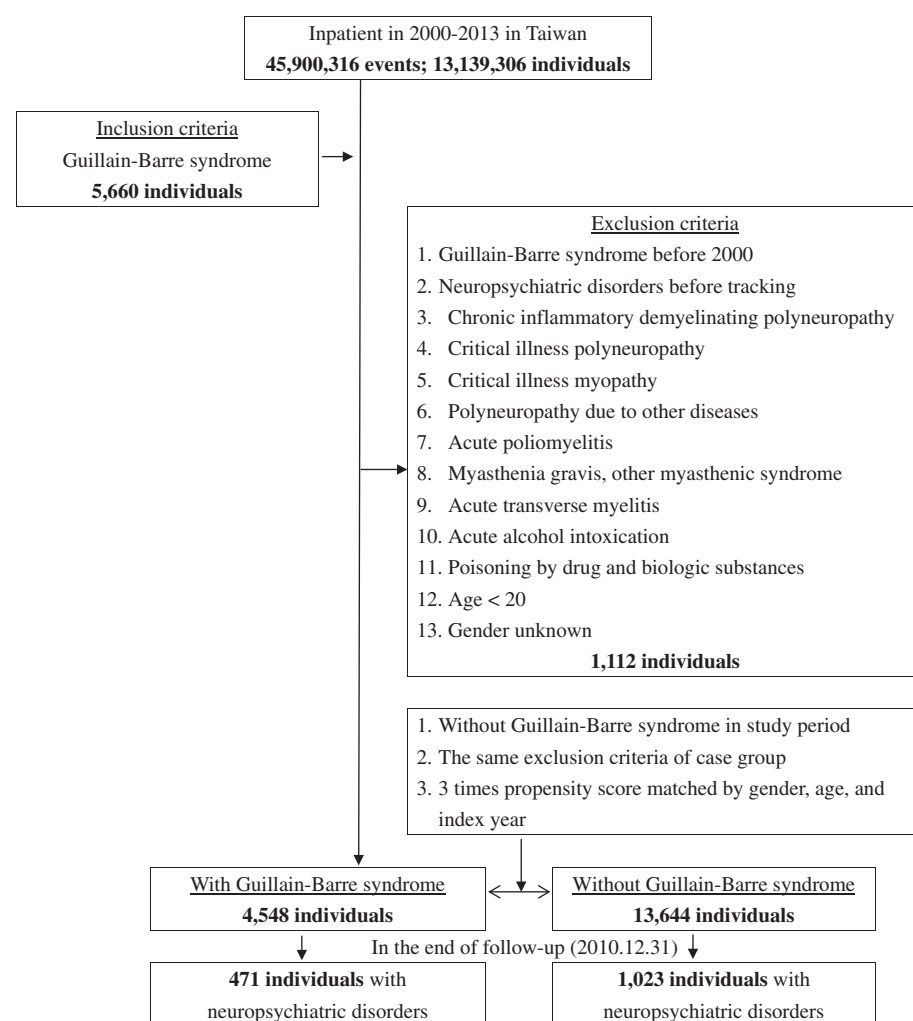


Fig. 1. The flowchart of study sample selection from National Health Insurance Research Database in Taiwan.

2.7. Statistical analysis

All statistical analyses were performed using the SPSS for Windows, version 22.0 (IBM Corp., Armonk, NY). χ^2 and t -tests were used to evaluate the distributions of the categorical and continuous variables, respectively, with a Fischer exact examination. Fine and Gray's competing risk analysis was used to determine the risk of psychiatric disorders since death can act as a competing risk factor for psychiatric disorders [36–38]. The results were presented as hazard ratio (HR) with a 95% confidence interval (CI). Differences in the risk of psychiatric disorders between the study and control groups were estimated using the Kaplan-Meier method with the log-rank test. A 2-tailed p value < 0.05 was considered to indicate statistical significance.

3. Results

3.1. Sample characteristics

Table 1 shows that the subject group had higher pulmonary complications, respiratory failure, systemic infection, and thrombotic complications. The study group also tended to search for medical help during the seasons of spring, summer, and autumn, live in northern Taiwan, and reside more in the regions of urbanization levels 1 and 2. The subject group also had a higher therapeutic plasma exchange, mechanical ventilation, and hemodialysis, as well as higher rates of treatment in medical centers and regional hospitals. There were no differences in the distribution of gender, age, CCI scores, and insurance

premiums between these two groups.

3.2. Kaplan-Meier model for the cumulative risk of psychiatric disorders

Of the total 18,192 enrollees, 4548 were the study subjects with GBS, and 13,644 were the 1:3 of the propensity score matched-controls. The 471 from the 4548 (10.36%) in the study subjects, in comparison to the 1023 from the 13,644 (7.50%), developed psychiatric disorders, and the difference was statistically significant in the Kaplan-Meier survival analysis (log-rank, $p < 0.001$, Fig. 2).

3.3. Hazard ratios analysis of psychiatric disorders in the patients with GBS

Supplementary Table 1 shows that in the competing analysis model, the hazard ratio of the study subjects in the development of psychiatric disorders was 4.320 (adjusted hazard ratio [HR], (95% CI: 3.852–4.842, $p < 0.001$), in comparison to the control group, after adjustment for gender, age, comorbidity (CCI scores), complications, residence/regions, and insurance premiums. The groups aged 30–59 were associated with a lower risk in comparison to the group aged 20–29. Besides, those who have had cardiac complications, systemic infection, and used mechanical ventilation or hemodialysis, were associated with a lower risk in developing psychiatric disorders. The GBS-group who had used therapeutic plasma exchange was associated with a 4.552-fold risk of developing psychiatric disorders. Subjects who lived in the residence of urbanization Levels 1 to 3 were associated with a lower risk of developing psychiatric disorders. Patients who were

Table 1
Characteristics of study at the baseline.

Guillain-Barre syndrome	With		Without		P
	n	%	n	%	
Variables					
Total	4548	25.00	13,644	75.00	
Gender					0.999
Male	2819	61.98	8457	61.98	
Female	1729	38.02	5187	38.02	
Age group (years)					0.999
20–29	482	10.60	1446	10.60	
30–39	521	11.46	1563	11.46	
40–49	656	14.42	1968	14.42	
50–59	760	16.71	2280	16.71	
60–69	896	19.70	2688	19.70	
≥70	1233	27.11	3699	27.11	
Charlson comorbidity index (CCI)	0.82 ± 1.69		0.84 ± 1.80		0.426
Cardiac complications					< 0.001
Without	4467	98.22	13,150	96.38	
With	81	1.78	494	3.62	
Pulmonary complications					< 0.001
Without	4226	92.92	13,531	99.17	
With	322	7.08	113	0.83	
Respiratory failure					< 0.001
Without	4364	95.95	13,611	99.76	
With	184	4.05	33	0.24	
Systemic infection					< 0.001
Without	4425	97.30	13,436	98.48	
With	123	2.70	208	1.52	
Thrombotic complications					0.013
Without	4543	99.89	13,642	99.99	
With	5	0.11	2	0.01	
Season					< 0.001
Spring (March–May)	1240	27.26	3677	26.95	
Summer (June–August)	1128	24.80	3345	24.52	
Autumn (September–November)	1081	23.77	2935	21.51	
Winter (December–February)	1099	24.16	3687	27.02	
Location					< 0.001
Northern Taiwan	2064	45.38	5344	39.17	
Middle Taiwan	1219	26.80	3704	27.15	
Southern Taiwan	1044	22.96	3692	27.06	
Eastern Taiwan	213	4.68	814	5.97	
Outlets islands	8	0.18	90	0.66	
Urbanization level					< 0.001
1 (The highest)	1792	39.40	4605	33.75	
2	2077	45.67	5759	42.21	
3	205	4.51	1068	7.83	
4 (The lowest)	474	10.42	2212	16.21	
Level of care					< 0.001
Hospital center	2572	56.55	4066	29.80	
Regional hospital	1593	35.03	3924	28.76	
Local hospital	383	8.42	5654	41.44	
Therapeutic plasma exchange					< 0.001
Without	4191	92.15	13,644	100.00	
With	357	7.85	0	0.00	
Mechanical ventilator					< 0.001
Without	4159	91.45	13,484	98.83	
With	389	8.55	160	1.17	
Hemodialysis					< 0.001
Without	3653	80.32	12,101	88.69	
With	895	19.68	1543	11.31	
Insured premium (NT\$)					0.867
< 18,000	4479	98.48	13,450	98.58	
18,000–34,999	53	1.17	146	1.07	
≥35,000	16	0.35	48	0.35	

P-value (category variable: Chi-square/Fisher exact test; continue variable: t-test).

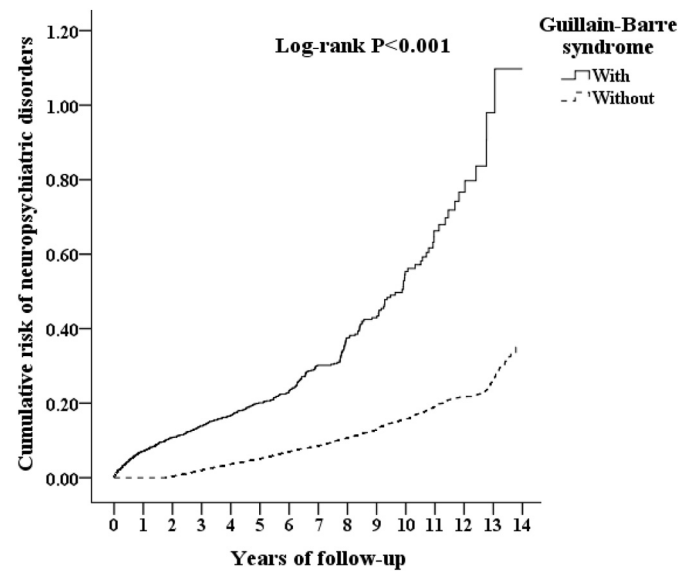


Fig. 2. Kaplan-Meier for cumulative risk of neuropsychiatric disorders among aged 20 and over stratified by Guillain-Barre syndrome with log-rank test.

treated in medical centers and regional hospitals also had a lower risk. In comparison to the patients who developed GBS during spring, the GBS patients during autumn had a lower risk of developing psychiatric disorders. In addition, in the study period, the average neurology outpatient visits in the GBS group were more than that in the non-GBS group (4.52 ± 5.49 vs 2.39 ± 3.95 visits, $p < 0.001$, data not shown), but the difference of the psychiatry visits were insignificant (2.12 ± 3.49 vs 2.23 ± 4.42 visits, $p = 0.126$, data not shown).

3.4. Hazard ratio analysis of psychiatric disorders in the patients with comorbidities and complications

Table 2 shows that the study group was associated with a higher risk of psychiatric disorders than the control group, regardless of gender, age, comorbidity (CCI scores), cardiac complications, pulmonary complications, respiratory failure, systemic infections, residence/regions, seasons, levels of hospitals, or insurance premiums in the subgroup analysis. The study subjects without thrombotic complications or therapeutic plasma exchanges were associated with a higher risk in developing psychiatric disorders.

3.5. Types of psychiatric disorders after GBS

Supplementary Table 2 reveals that the study subjects were associated with an increased risk of developing dementia, depressive disorders, sleep disorders, and psychotic disorders, with an adjusted HR as 3.992 ($p = 0.035$), 4.834 ($p < 0.001$), 3.906 ($p < 0.001$), and 2.628 ($p = 0.047$). In addition, of the total 4548 GBS subjects in this dataset, the prevalence of these disorders were 4.02%, 2.57%, 2.68%, and 0.64%, respectively, in this 13-year study. However, no subjects with PTSD or acute stress disorder were found.

4. Discussion

4.1. Association between GBS and the risk of psychiatric disorders

In this study, we found that the adult patients with GBS had an increased risk of neuropsychiatric disorders. Even after adjusting for covariates, the overall adjusted HR was 4.320 (95% CI: 3.852–4.842, $p < 0.001$). In other words, the adult patients with GBS had a 4.3-fold increased risk of developing psychiatric disorders. The Kaplan-Meier analysis revealed that the study subjects had a significantly higher 13-

Table 2

Factors of neuropsychiatric disorders at the end of follow-up stratified by variables listed in the table by using Cox regression and Fine and Gray's competing risk model.

Guillain-Barre syndrome Variables	No competing risk in the model				Competing risk in the model			
	Adjusted HR	95%CI	95%CI	P	Adjusted HR	95%CI	95%CI	P
Total	4.336	3.868	4.860	< 0.001	4.320	3.853	4.842	< 0.001
Gender								
Male	4.308	3.724	4.984	< 0.001	4.272	3.693	4.934	< 0.001
Female	4.339	3.602	5.228	< 0.001	4.349	3.608	5.242	< 0.001
Age group (years)								
20–29	6.995	3.010	16.258	< 0.001	7.003	3.014	16.276	< 0.001
30–39	7.452	4.762	11.683	< 0.001	7.316	4.681	11.433	< 0.001
40–49	8.437	5.755	12.369	< 0.001	8.406	5.722	12.350	< 0.001
50–59	4.610	3.247	6.546	< 0.001	4.620	3.251	6.567	< 0.001
60–69	5.522	4.224	7.218	< 0.001	5.514	4.217	7.210	< 0.001
≥ 70	3.201	2.714	3.776	< 0.001	3.190	2.704	3.763	< 0.001
Cardiac complications								
Without	4.369	3.887	4.910	< 0.001	4.352	3.872	4.892	< 0.001
With	6.982	4.103	11.883	< 0.001	7.023	4.128	11.950	< 0.001
Pulmonary complications								
Without	4.166	3.708	4.683	< 0.001	4.149	3.691	4.685	< 0.001
With	11.634	8.057	22.345	< 0.001	12.817	6.572	24.994	< 0.001
Respiratory failure								
Without	4.229	3.767	4.747	< 0.001	4.204	3.745	4.720	< 0.001
With	21.419	8.714	52.643	< 0.001	26.375	10.387	66.973	< 0.001
Systemic infection								
Without	4.244	3.777	4.767	< 0.001	4.230	3.765	4.753	< 0.001
With	58.743	23.951	#####	< 0.001	58.523	23.687	#####	< 0.001
Thrombotic complications								
Without	4.336	3.868	4.860	< 0.001	4.320	3.853	4.842	< 0.001
With	–	–	–	–	–	–	–	–
Season								
Spring (March–May)	4.242	3.334	5.363	< 0.001	4.301	3.381	5.472	< 0.001
Summer (June–August)	4.010	3.183	5.053	< 0.001	3.953	3.137	4.982	< 0.001
Autumn (September–November)	4.946	3.967	6.176	< 0.001	4.908	3.936	6.119	< 0.001
Winter (December–February)	4.201	3.343	5.280	< 0.001	4.153	3.303	5.221	< 0.001
Urbanization level								
1 (The highest)	5.562	4.519	7.070	< 0.001	5.759	4.599	7.213	< 0.001
2	4.263	3.600	5.048	< 0.001	4.229	3.571	5.009	< 0.001
3	3.823	2.452	5.958	< 0.001	3.745	2.403	5.836	< 0.001
4 (The lowest)	3.650	2.812	4.739	< 0.001	3.629	2.796	4.710	< 0.001
Level of care								
Hospital center	5.214	4.247	6.402	< 0.001	5.159	4.202	6.334	< 0.001
Regional hospital	4.215	3.522	5.044	< 0.001	4.303	3.593	5.152	< 0.001
Local hospital	3.736	2.982	4.681	< 0.001	3.671	2.930	4.601	< 0.001
Therapeutic plasma exchange								
Without	4.333	3.865	4.857	< 0.001	4.314	3.848	4.836	< 0.001
With	–	–	–	–	–	–	–	–
Mechanical ventilator								
Without	4.160	3.695	4.684	< 0.001	4.144	3.680	4.666	< 0.001
With	7.997	5.063	12.632	< 0.001	8.062	5.101	12.742	< 0.001
Hemodialysis								
Without	4.246	3.760	4.794	< 0.001	4.223	3.740	4.769	< 0.001
With	5.125	3.623	7.251	< 0.001	5.189	3.666	7.343	< 0.001
Insured premium (NT\$)								
< 18,000	4.347	3.875	4.876	< 0.001	4.321	3.852	4.847	< 0.001
18,000–34,999	3.274	0.846	12.670	0.086	4.625	1.072	19.952	0.040
≥ 35,000	–	–	–	–	–	–	–	–

PYs = Person-years; Adjusted HR = Adjusted Hazard ratio; Adjusted for all the variables listed in this table; CI = confidence interval.

year psychiatric disorders-free survival rate than the controls. The younger GBS subjects aged < 40 were considered to be at a lower risk of developing neuropsychiatric disorders. This might be the first study on the risk of psychiatric disorders in GBS patients using a nationwide, population-based dataset.

4.2. Comparison of this study to previous literatures

In one study of 139 patients with GBS in an intensive care unit, 31% experienced mental status changes, which included vivid dreams (19%), illusions (30%), hallucinations (60%, mainly visual), and delusions (70%, mostly paranoid), within a mean of nine days after the onset of the disease, and lasted a median of eight days [13]. Two previous studies found that approximately 50% of the GBS patients

suffered from sleep problems during their hospital stay [14,15]. In our study, we only enrolled patients with three visits for their psychiatric diagnoses, and this might be the reason of the relatively low prevalence of sleep disorder (2.68%), and psychotic disorder (0.64%), in the GBS-group. Furthermore, the strength of our study is in the longer duration of follow up of the psychiatric disorders in the GBS patients after their acute stage or hospital stay. This is a reminder for the clinicians that careful follow up of the mental health in patients after recovery from GBS is of the utmost importance.

4.3. Possible mechanisms for the increased risk of psychiatric disorders in patients with GBS

The underlying mechanisms of the increased risk of psychiatric disorders in the GBS-subjects remained unclear. Traumatic stress from a life-threatening illness might be one of the contributory factors [8,9]. Long-term disabilities usually exist in cases of GBS, and psychiatric disorders can occur during the recovery phase following the acute phase [5,6]. In our study, the patients who had received therapeutic plasma exchanges had a higher risk as per the adjusted HR of 4.552 (95% CI: 1.850–11.142, $p < 0.001$), and this might hint that the severity of GBS could be associated with more psychiatric morbidity, even though other treatments such as hemodialysis and mechanical ventilation were associated with a lower risk of psychiatric disorders.

A Taiwan study showed that, in spring, the incidence was 10% higher than in other seasons [3], and our study found that the patients who suffered GBS in the autumn had a slightly lower (0.732-fold, $p < 0.001$) risk of psychiatric disorders in comparison to those in the spring. However, the association between seasonal factors and the subsequent risk of psychiatric disorders in GBS patients is needed for future study.

In GBS, neuroinflammation is mainly involved in the peripheral nervous system, but studies have found that the central nervous system involvement in GBS patients, such as mononuclear inflammatory lesions, and the microglial activation in conjunction with the secondary myelin impairment, were observed in the spinal cord, medulla oblongata, pons, and midbrain of the GBS autopsy cases, and these findings might well hint that the CNS involvement may also be important to other psychiatric morbidity [39,40]. Furthermore, GBS could be associated with the elevation of the T helper (Th) 17 cells and their cytokines [41,42]. We may well hypothesize that such pro-inflammatory factors might also contribute to the pathogenesis of dementia that have been found in other studies [43–45]. Besides, therapeutic agents such as corticosteroids might also be associated with psychiatric disorders [46]. The bidirectional relationship between depression and immunity could also contribute to the psychiatric morbidity [47,48]. The unknown reason of the lower CSF hypocretin-1 levels had also been found to be related to psychosis in the acute phase [13], but the long term effects remained unclear.

4.4. Limitations

The present study has several limitations that warrant consideration. First, similar to the previous studies using the NHIRD on GBS [3,4,49], not all data were recorded in the NHIRD, and we were unable to evaluate the neurological severity, weakness severity, laboratory parameters, the availability of rehabilitation, and the additional examination findings (e.g., electrophysiological testing). Second, other factors, such as genetic, psychosocial, and environmental factors, were not included in the dataset. However, since the present study had covered all of Taiwan's hospitals and > 99% of the Taiwanese population during a 13-year period, which thereby increased the likelihood of our data being valid and representative. Furthermore, in this study, patients with a history of GBS have visited the neurology department more frequently than patients without a history of GBS, thus, leading to the potentially more frequent psychiatric referrals and diagnosis.

However, the difference of psychiatry outpatient visits was insignificant between the two groups.

5. Conclusions

This study has shown evidence to support that the patients with GBS have a 4.3-fold risk of developing neuropsychiatric disorders. Other large or national data sets should be completed to confirm this finding and to explore these underlying mechanisms. This finding should, indeed, be a reminder to clinicians with regard to the risk of subsequent psychiatric disorders. Even though the careful follow up of the mental health in patients is crucial, their recovery from GBS is very important.

Supplementary data to this article can be found online at <http://dx.doi.org/10.1016/j.jns.2017.08.022>.

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